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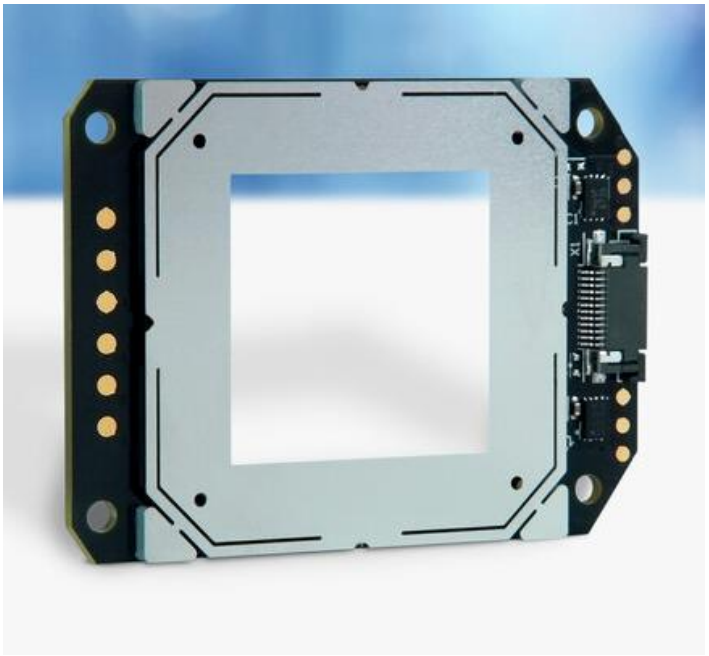

Extended Pixel Resolution Actuator XPR-20®

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1. Overview

The Optotune XPR-20 is an Extended Pixel Resolution 4-position Actuator. It has been designed to provide fast beam shifting to enable resolution enhancement of imaging and projection systems.

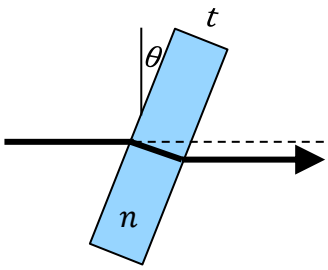
Main specifications

Clear aperture	21 x 21 (20 x 20 usable)	mm
Size (width x height x depth)	47 x 35 x 3.3, max height 3.65	mm
Positions	4	
Glass movement full angle	0.225 in x and y	Degree
Pixel shift	2.7 in x and y	µm
Movement angle / pixel shift tolerance[1]	Typical: <±5% Maximal: 12.5%	
Frequency	50 (±0.5) and 60 (±0.5)	Hz
Sound pressure*	Typical: <20; Max: 23 in DAC mode (at 0.3m) Typical: <20; Max: 25 in PWM mode (at 1m)	dBA
Weight	9	g
Drive current (bi-directional)	Average: <±200	mA
Power consumption (average)	< 1.5	W

* see Section 6 for the details and measurement conditions.

Working principle

The image is shifted laterally by tilting a planar glass window. The relation between displacement and angle can be seen in Figure 1.



$$\Delta y = t\,sin\theta\left(1-\sqrt{\frac{1-sin^2\theta}{n^2-sin^2\theta}}\right)$$

Figure 1: Tilting a glass window results in a lateral image shift.

The XPR-20 tilts a glass window in two axes to reach four distinct positions. As a result, each pixel is projected at four locations increasing the resolution by a factor of four.

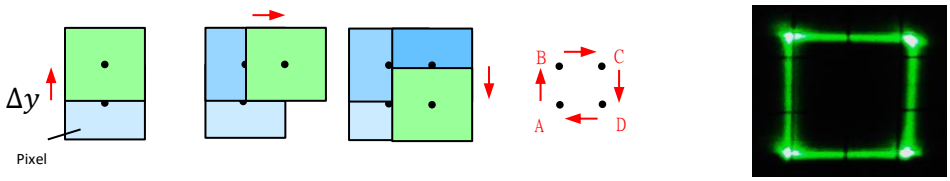


Figure 2: Explanation of 4-position (4P) actuator principle

[1] Measured for each device.

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Pointing accuracy and transition time

The below graph shows the optical performance of the XPR-20 working with a nominal 50Hz waveform optimized for 1.2ms transition time. Orange line shows the X-axis and blue line the Y-axis.

The main criteria are the accuracy of the angular position and the speed of the transition between the different positions. The angular position during a steady state should not exceed the ±12.5% of nominal angular distance (thin grey lines) and the average angular position during the steady state do not exceed the ±5% of the nominal angular distance (interval indicated by thick black lines). The transition time is defined as the time it takes to switch from one ±12.5% "tolerance line" to the next.

With different input waveforms the behaviour can be tuned for different movement patterns. It can be optimized for short transition times or for very stable flat top profiles.

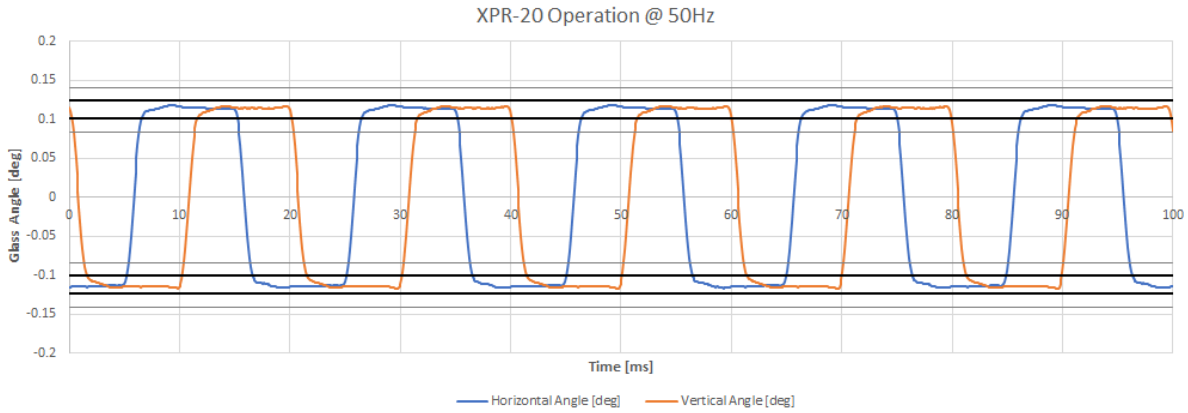

Figure 3: Typical response at 50Hz; transition time 1.2ms. Grey lines (thinner lines) define the allowed ±12.5% region for max. oscillation, while the thicker black line indicates the tolerance for average steady-state angle.

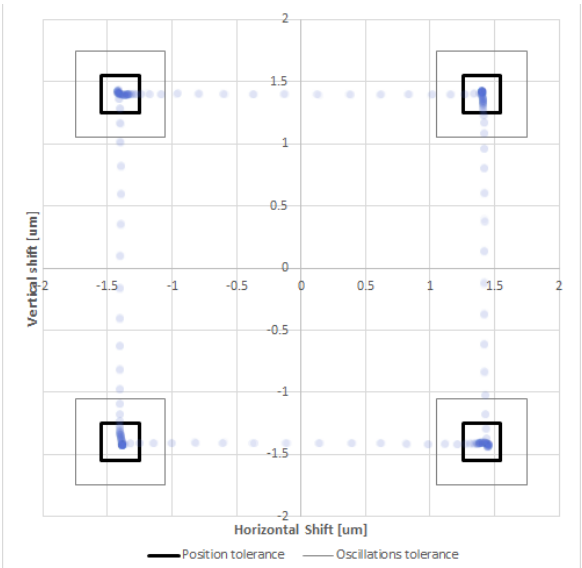

Figure 4: Typical pixel shift trajectory: Grey lines (thinner lines) define the allowed ±12.5% region for maximum oscillation, while the thicker black line indicates the tolerance for average steady-state angle.

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2. Glass Specifications

Glass Specifications

Material	B270	
Refractive index @590nm	1.523	
Abbe Number (Vd)	58.5	
Thickness	2.0 ±0.05	mm
Surface Flatness	3/4(2) according to ISO 10110	
Surface Defects	5/ 5x0.2; L1x0.04; C3x0.2 according to ISO 10110	
Transmission	>98% from 400 – 680nm, AOI 0…34°	
Optical Flux max thermal load optical power (CW, 400…700nm)	 10.64 300	 W/cm^2

Transmission range

The glass of the actuator is AR coated on both sides to achieve >99% average transmission from 400 – 680nm over a wide angle.

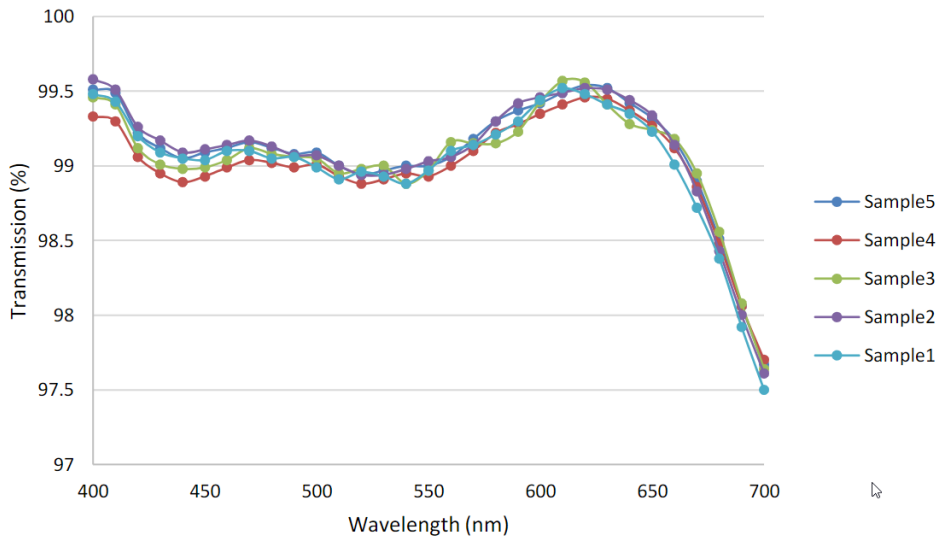

Figure 5: Transmission spectrum of XPR-20 glass with VIS coating (5 samples measured at 0° incident angle)

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3. Environmental Specifications

Environmental specifications

P/N	142-000-02	142-000-03	142-000-04	142-000-05	
Operating conditions					
Environment Temperature *	0 – 75				°C
Humidity	0 – 85, non-condensing				%RH
Max. dew point	40				°C
Altitude	<12,000 / 4000				ft / m
Storage conditions					
Temperature	-40 – 85				°C
Temperature shipping tray	-40 – 70				°C
Humidity	0 – 85, non-condensing				%RH
Max. dew point	50				°C
Altitude	<40,000 / 13000				ft / m
Mechanical shock	105G with 15ms half-sine				
Lifetime in operation (survival)	20,000				h
Performance after 10,000h					
Movement angle/ Pixel shift tolerance	Typical: +/-7% Max: 15%				
Transition time (typical/ maximum)	1.3 / 1.5	1.4 / 1.6	1.3 / 1.5	1.4 / 1.6	ms
Compliance	CE, RoHS, REACH, flammability UL94 V-0				

* specific sections of the XPR can/will get warmer but will stay in their material specifications within this environment temperature range. Please refer to Section 6 for details.

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4. Mechanical Specifications

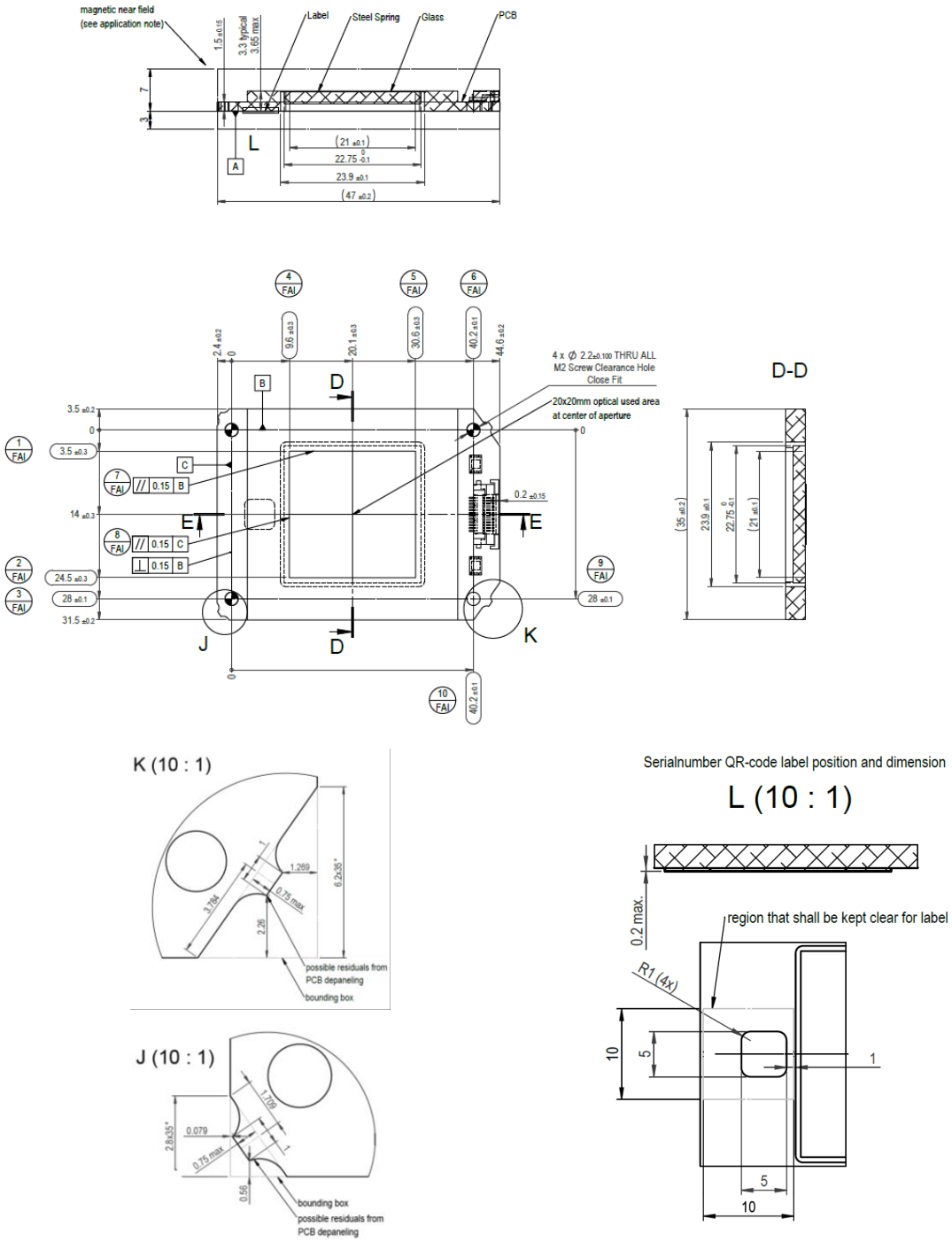

Figure 6: Dimensions of the XPR-20[2];
Check dimensions are marked by a number (1 to 10)

[2] Residuals from manufacturing (Details J and *K*) can vary but stay within the bounding box.

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Mounting Instructions

- A minimum clearance of 0.3mm from the top surface (metal sheet) must be guaranteed.
- For optimum performance, the XPR-20 shall be mounted on a solid base with 4 x M2 screws with washers of max 5mm diameter.
- The screws shall be tightened w/ a torque of 10±2Ncm.
- The mounting surface shall be flat ±0.1mm.

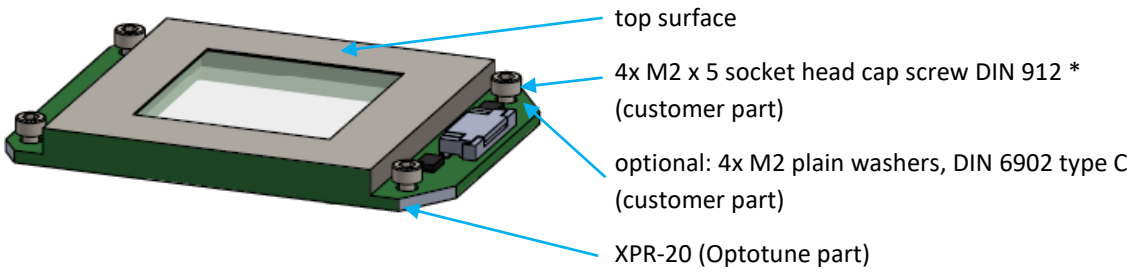

Figure 7: Mounting of the XPR-20

*) other head forms, such as flat head or longer thread length are also possible and do not affect the performance of the XPR-20.

The XPR-20 contains strong magnets needed for efficient actuation. In the magnetic near field of the XPR-20, as shown in Figure 8 there must NOT be a significant amount of ferromagnetic material or magnets. This explicitly includes stainless steel, which can have magnetic properties. Small parts, such as stainless-steel screws are acceptable. If in doubt, please contact Optotune.

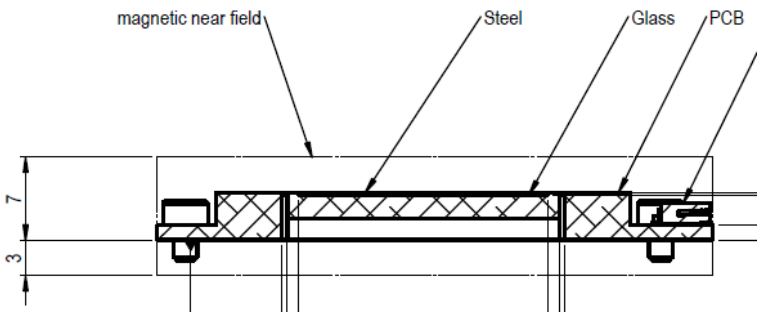

Figure 8: Definition of the magnetic near field of the XPR-20

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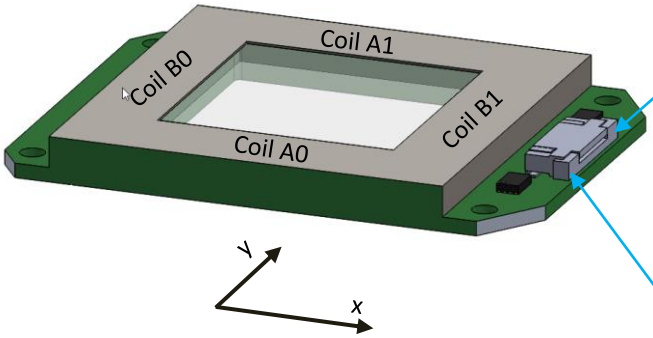
5. Electrical specifications

Pin Assignment

The XPR-20 provides electrical connection through a 12-pin connector (Amphenol SFV12R-2STE1HLF).



Pin-out:

P12	N.C.
P11	N.C.
P10	N.C.
P09	N.C.
P08	GND
P07	I2C_SCL
P06	I2C_SDA
P05	Vcc (3.3V)
P04	Axis Y-
P03	Axis Y+
P02	Axis X-
P01	Axis X+

Figure 9: Pin assignment of the XPR-20

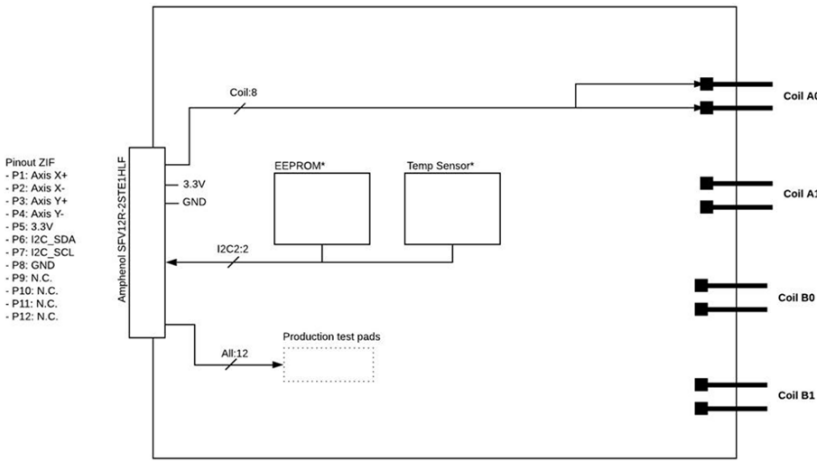


Figure 10: Block diagram of the XPR-20

Actuator motors

Number of motors	2 per channel, 2 channels,	4	
Resistance per channel	Typical, 4000ppm/K	9.8	Ω
Max. supply current (pulse)	During a pulse of <2ms at 120Hz	600	mA
Max. supply current (rms value)		300	mA

Attention: Exceeding the maximum supply current can destroy the motor

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Electronic Components

Temperature Sensor

Number of temp. sensors	LM75BTP,147 or compatible*	-	1
Supply voltage	Nom.	V	3.3
Max. supply current	Bus active, 400kHz	uA	300
Interface		-	I2C
Two-wire address		-	1001100x (R: 0x98 / W: 0x99)
Max. conversion rate		Hz	10
Min. SCL operational frequency		kHz	0.02
Max. SCL operational frequency		kHz	400
Min. sensing temperature		°C	-55
Max. sensing temperature		°C	125
Resolution	Typ.	°C	0.125
Accuracy	Typ., -25 … 100°C Typ., -55 … 125°C	°C	±2 ±3

EEPROM

Number of EEPROM chips	M24C64-FMC6TG or compatible*	-	1
Supply voltage	Nom.	V	3.3
Max. read current		mA	2
Interface		-	I2C
Two-wire address		-	1010000x (R: 0xA0 / W: 0xA1)
Storage size of header	Version, S/N, Part Conf, Operation specs	Byte	32
Storage size of data sets		Byte	256
Storage size of ET		Byte	4160
Optotune information block		Byte	32
Min. Storage Needed		Byte	4480
Tot. EEPROM space		Byte	8K

* compatible component lists are available upon request.

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6. Thermal Specifications

The XPR-20 contains a build-in temperature sensor to monitor the device temperature and to allow the controller to adjust for thermal drifts, as shown in Figure 11

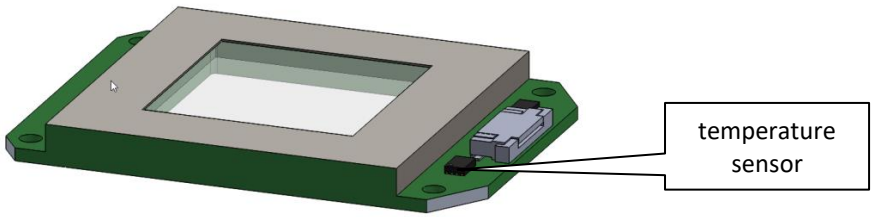
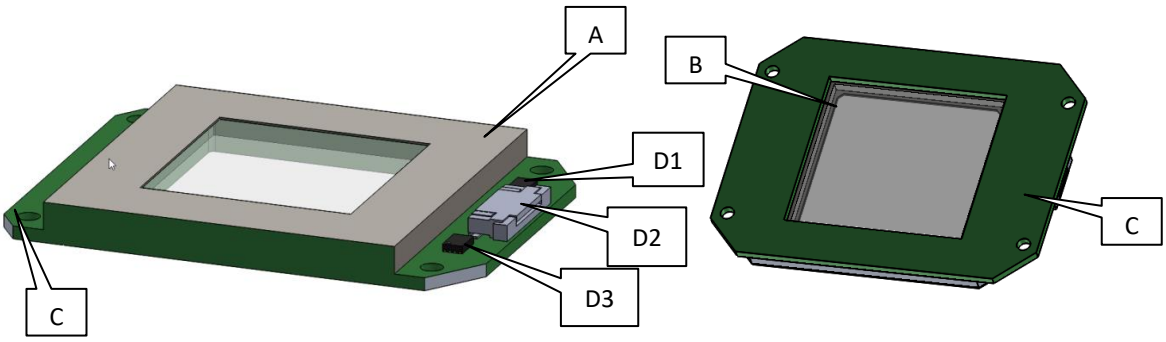


Figure 11: Location of the built-in temperature sensor on the XPR-20

For reliable operation within specifications, the following temperature values on the actuator must not be exceeded in operation:



Component	Temperature Limit (°C)
A: spring	140
B: glass gluing area	120
C: PCB	110
D: connector + electronic components	85[3]

Figure 12: Temperature Limits of the XPR-20

7. Safety and compliance

The product fulfils the CE, RoHS, REACH and flammability UV94 V-0 compliance standards.

The customer is solely responsible to comply with all relevant safety regulations for integration as well as EMC standards for integration and operation.

[3] Temperature limit shows the safe value that the thermal couples shall measure. Components in D region: D1 (EEPROM): ambient operating temp up to 85 °C ; D2(connector): ambient temp up to 105 °C; D3(temp sensor): ambient temp up to 125 °C

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